Homework 2: Operations on Signals
CPE 381

Instructor: Rahul Bhadani

Due: Sept 27, 2026, 11:59 PM
116 points

You are allowed to use a generative model-based AI tool for your assignment. However, you must submit an accompanying reflection report detailing how you used the AI tool, the specific query you made, and how it improved your understanding of the subject. You are also required to submit screenshots of your conversation with any large language model (LLM) or equivalent conversational AI, clearly showing the prompts and your login avatar. Some conversational AIs provide a way to share a conversation link, and such a link is desirable for authenticity. Failure to do so may result in actions taken in compliance with the plagiarism policy.

Additionally, you must include your thoughts on how you would approach the assignment if such a tool were not available. Failure to provide a reflection report for every assignment where an AI tool is used may result in a penalty, and subsequent actions will be taken in line with the plagiarism policy.

Submission instruction:

Upload a .pdf on Canvas with the format `CPE381-LastFirst-HW-XX`. For example, if your name is Sam Wells, your file name should be `CPE381-WellsSam-HW-XX.pdf`. If there is a programming assignment, then you should include your source code along with your PDF files in a zip file `CPE381-LastFirst-HW-XX.zip`. Your submission must contain your name and UAH Charger ID or your UAH email address. Please number your pages as well.

Theory

1 Linear Interpolation (6 Points)

Assume three signal values $x = \{3.0, 4.0, 3.5\}$ at timestamp $t = \{2.0, 4.0, 6.0\}$. We will consider the case of linear interpolation.

1. How many linear segments are required if given three samples or signal values? **(2 Points)**

2. Write down all the equation of line segments necessary for linear interpolation given the three samples above. **(2 Points)**

3. What is the value of when interpolated at $t = 2.5$? **(2 Points)**

Answer

1. 2

2. The Equation of line connecting two points are

$$\frac{x_2 - x_1}{t_2 - t_1} = \frac{x - x_1}{t - t_1} \tag{1}$$

The equation of the first line segment is :

$$\begin{aligned} \frac{4-3}{4-2} &= \frac{x-3}{t-2} \\ \frac{1}{2} &= \frac{x-3}{t-2} \\ t-2 &= 2x-6 \\ t &= 2x-4 \\ x &= \frac{1}{2}(t+4) \end{aligned} \tag{2}$$

The equation of the second line segment is:

$$\begin{aligned} \frac{3.5-4}{6-4} &= \frac{x-4}{t-4} \\ \frac{-0.5}{2} &= \frac{x-4}{t-4} \\ -(t-4) &= 4x-16 \\ x &= \frac{1}{4}(20-t) \end{aligned} \tag{3}$$

3. Since $t = 2.5$ is between 2.0 and 4.0 we will plug its value into the first segment.

$$x = 0.5 * (2.5 + 4) = 3.25 \tag{4}$$

2 Cubic Spline Interpolation (10 Points)

Consider signal values $x = \{0.0, 7.0, 9.0, 3.0, 4.0\}$ at timestamp $t = \{0.0, 1.0, 2.0, 4.0, 5.0\}$. We will consider the case of cubic spline interpolation.

1. Assuming free boundary condition, write the B matrix corresponding to all the b_i coefficients as per formulation provided in Chapter 02 Slides. **(6 Points)**
2. What are the values of a_1, b_1, c_1, d_1 following the convention for cubic spline interpolation formulation as described in the Chapter 02 slides? **(4 Points)**

Answer

1. In the case of free boundary condition, $b_1 = 0$, $b_5 = 0$.

   And

$$\begin{aligned} h_1 &= t_2 - t_1 = 1 - 0 = 1 \\ h_2 &= t_3 - t_2 = 2 - 1 = 1 \\ h_3 &= t_4 - t_3 = 4 - 2 = 2 \\ h_4 &= t_5 - t_4 = 5 - 4 = 1 \end{aligned} \tag{5}$$

   From the lecture slides, the formula for b_2 to b_n in matrix form is

$$\begin{bmatrix} 2\cdot(1+1) & 1 & 0 \\ 1 & 2\cdot(1+2) & 2 \\ 0 & 2 & 2\cdot(2+1) \end{bmatrix} = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 6 & 2 \\ 0 & 2 & 6 \end{bmatrix} \tag{6}$$

   If we include b_1 and b_5 in the systems of equations, B matrix can be written as

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 2\cdot(1+1) & 1 & 0 & 0 \\ 0 & 1 & 2\cdot(1+2) & 2 & 0 \\ 0 & 0 & 2 & 2\cdot(2+1) & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 4 & 1 & 0 & 0 \\ 0 & 1 & 6 & 2 & 0 \\ 0 & 0 & 2 & 6 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{7}$$

   The full systems of equations for b_2 to b_4 is given by

$$\begin{bmatrix} 4 & 1 & 0 \\ 1 & 6 & 2 \\ 0 & 2 & 6 \end{bmatrix} \begin{bmatrix} b_2 \\ b_3 \\ b_4 \end{bmatrix} = \begin{bmatrix} \frac{3(d_3-d_2)}{h_2} - \frac{3(d_2-d_1)}{h_1} \\ \frac{3(d_4-d_3)}{h_3} - \frac{3(d_3-d_2)}{h_2} \\ \frac{3(d_5-d_4)}{h_4} - \frac{3(d_4-d_3)}{h_3} \end{bmatrix} = \begin{bmatrix} \frac{3(9-7)}{1} - \frac{3(7-0)}{1} \\ \frac{3(3-9)}{2} - \frac{3(9-7)}{1} \\ \frac{3(4-3)}{1} - \frac{3(3-9)}{2} \end{bmatrix} = \begin{bmatrix} -15 \\ -15 \\ 12 \end{bmatrix} \tag{8}$$

   which upon solving using MATLAB gives $b_2 = -3$, $b_3 = -3$, and $b_4 = 3.0$.

2. $d_1 = x_1 = 0.0$.

   $c_1 = \dfrac{d_2 - d_1}{h_1} - \dfrac{1}{3}(2b_1 + b_2)h_1 = \dfrac{7-0}{1} - \dfrac{1}{3}(-3 \cdot 1) = 7 + 1 = 8$

   $a_1 = \frac{b_2 - b_1}{3h_1} = \frac{-3-0}{3 \cdot 1} = -1$

   and because we assumed free boundary conditions $b_1 = 0$.

3 Play with Signals. (10 points)

A continuous-time signal x(t) is shown in Figure 1.

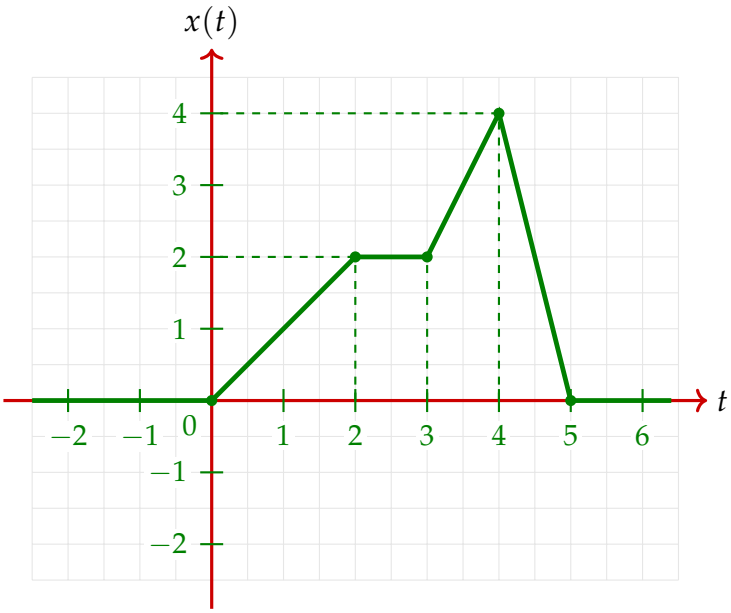


Figure 1: The ramp rises with unit slope until $t = 2$, holds the value 2 over $2 \leq t \leq 3$, climbs with slope 2 to its peak of 4 at $t = 4$, and falls to zero at $t = 5$.

Based on $x(t)$ plot, sketch and label each of the following signals: (i) $x(t-2)$; (ii) $x(2t)$; (iii) $x(t/2)$; (iv) $x(-t)$; (v) $-x(-t)$.

3.1 Answers:

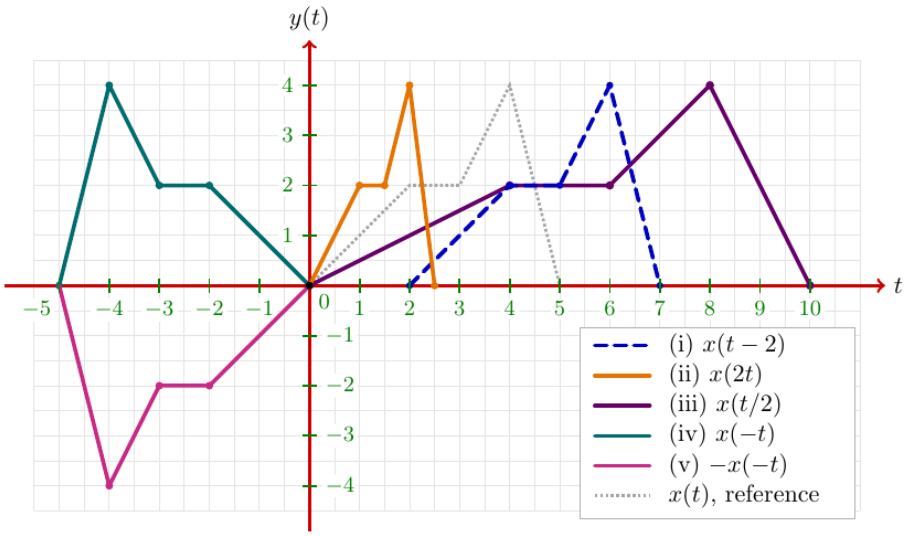


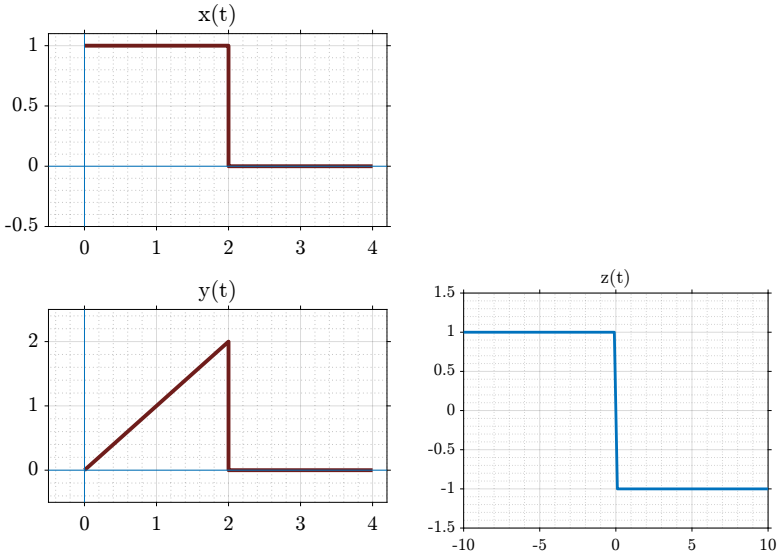
Figure 2: **Play with Signals** Answer

4 Picasso of Signals (15 points)

Sketch the following signals:

1. $x(t) = u(t) - u(t-a), \quad a > 0$
2. $y(t) = t[u(t) - u(t-a)], a > 0$
3. $z(t) = u(-t) - u(t)$ where $u(t)$ is the unit-step signal.

Answer



5 Divide and Conquer (10 points)

Determine the even and odd components of the following signals:

1. $x(t) = u(t)$.
2. $x(t) = \sin\left(\omega_0 t + \frac{\pi}{4}\right)$

Answer

1. $x_e(t) = \frac{1}{2}$, $x_o(t) = \frac{1}{2}\text{sgn}(t)$
2. $x_e(t) = \frac{1}{\sqrt{2}}\cos(\omega_0 t)$, $x_o(t) = \frac{1}{\sqrt{2}}\sin(\omega_0 t)$

6 Rinse and Repeat (15 points)

Determine whether or not each of the following signals is periodic. If signals are periodic, find their fundamental period.

1. $x(t) = \cos\left(2t + \frac{\pi}{4}\right)$.
2. $x(t) = \cos^2(t)$.

3. $x(t) = \cos(2\pi t)u(t)$.

Answer

1. Periodic, period $= \pi$.
2. Periodic, period $= \pi$.
3. Nonperiodic.

Practice

7 MATLAB Programming: Signal Plotting (10 points)

Consider

$$y(t) = A\cos(\Omega_c t + s(t))$$

1. $A = 1$, $\Omega_c = 2$, $s(t) = \frac{t^2}{4}$. Use MATLAB to plot the signal for $0 \leq t \leq 40$ seconds in the steps of 0.05. Use `sound` function to listen to the signal.
2. $A = 1$, $\Omega_c = 2$, $s(t) = -2\sin(t)$. Use MATLAB to plot the signal for $0 \leq t \leq 40$ seconds in the steps of 0.05.

7.1 Answers:

```
t=0:0.05:40;

y=cos(2*t+t.^2/4);
y1=cos(2*t- 2*sin(t));
fig = figure(14)
subplot(211)
plot(t,y); title('Part 1: linear chirp')
axis([0 20 1.1*min(y) 1.1*max(y)]);grid
subplot(212)
plot(t,y1);title('Part 2: sinusoidal chirp');xlabel('t')
axis([0 20 1.1*min(y1) 1.1*max(y1)]);grid
```

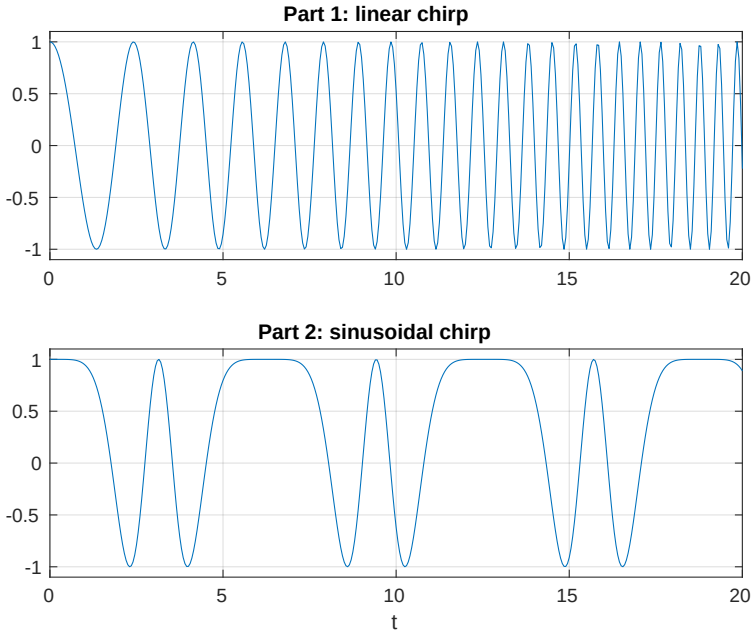


Figure 3: MATLAB Programming: Signal Plotting

8 Compact Form (20 points)

For Figure 4, represent the signal (f) in compact form using transformed version of unit-step function $u(t)$ and ramp signal $r(t)$ as necessary.

Write a MATLAB script to produce Figure 4. Note that the signal for $t < 0$ is zero.

Tip: You may use `heaviside` function in MATLAB to implement $u(t)$.

8.1 Answers:

Option 1:

```
% Define the time vector with a step of 0.01
t = 0:0.01:20;

% Initialize the function values
f = zeros(size(t));

% Define the piecewise function
for i = 1:length(t)
    if t(i) > 0 && t(i) <= 4
        f(i) = 2;
    elseif t(i) > 4 && t(i) <= 6
        f(i) = t(i);
    elseif t(i) > 6
```

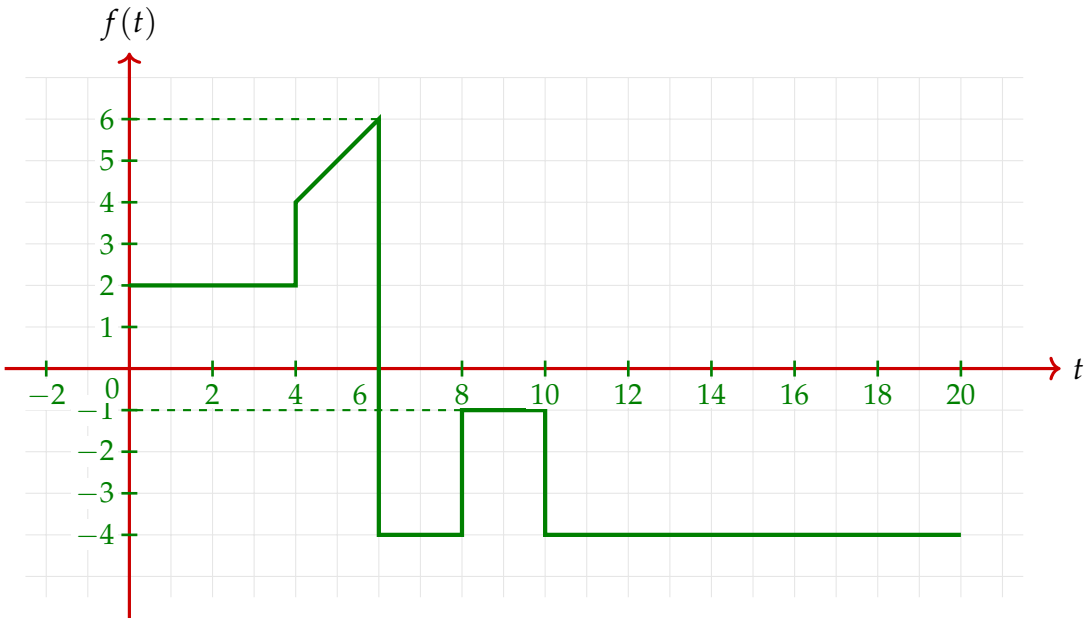


Figure 4: Compact Form

```
        f(i) = -4;
    end
end

% Create the plot
fig = figure;
fig.Position= [421         632        1139         606];
plot(t, f, 'LineWidth', 2);
xlabel('t', 'FontSize', 24, 'Interpreter', 'latex');
ylabel('f(t)', 'FontSize', 24, 'Interpreter', 'latex');
title('HW02 Q6 CPE381 FA24', 'FontSize', 24, 'Interpreter', 'latex');
grid on;
axis equal
set(gca, 'FontSize', 16);
```

Option 2:

```
%% Alternatively

% Define the time vector with a step of 0.01
t = 0:0.01:20;

% Define the piecewise function using unit-step (Heaviside) and ramp functions (t)
f = 2 * (heaviside(t) - heaviside(t - 4)) + (t) .* (heaviside(t - 4) -
```

```
        heaviside(t - 6)) - 4 * heaviside(t - 6);

% Create the plot
fig = figure;
fig.Position= [421         632        1139         606];
plot(t, f, 'LineWidth', 2);
xlabel('t', 'FontSize', 24, 'Interpreter', 'latex');
ylabel('f(t)', 'FontSize', 24, 'Interpreter', 'latex');
title('HW02 Q6 CPE381 FA24', 'FontSize', 24, 'Interpreter', 'latex');
grid on;
axis equal;
set(gca, 'FontSize', 16);
```

9 Cubic Spline Interpolation and Data-fitting (20 Points)

Consider renewable energy based electric grid and weather data for 1 week starting from 1/1/2018 12:00:00 AM in the zone 1 of ERCOT (Electric Reliability Council of Texas) that operates Texas's electrical grid, the Texas Interconnection. Data can be downloaded from https://github.com/rahulbhadani/CPE381_FA26/blob/main/Data/week_data_ERCOT_zone_1_.csv.

For those who are interest, the original source of data is https://doi.org/10.5281/zenodo.5130611.

The `time` column of the given CSV file is of the type `datetime`. `datetime` can be converted to POSIX format using

```
csv_file = 'week_data_ERCOT_zone_1_.csv';
week_data = readtable(csv_file);
px_time = posixtime(week_data.time);
```

The first entry, of the `px_time` is 1514764800.

You can substract the entire time column from the first value to make the first time value 0:

```
px_time = px_time - px_time(1);
```

For the give time series data, perform the following task:

1. Plot a line graph of the load power from the given data against `px_time`. Provide the figure as a part of your submission as well. **(5 Points)**
2. As the given data has time-granularity of 1 minute, the sampling time $\Delta t = 60\ s$, we are interested in resampling the data at $\Delta t = 1\ s$.

Write a MATLAB script to resample the load power data at $\Delta t = 1\ s$ using cubic spline interpolation technique. Use free boundary condition. You may use the cubic spline interpolation function implementation provided as a part of the course GitHub repository. **(10 Points)**

3. Plot the original load power data and resampled load power data together. Set x-axis limit to $[7100, 7300]$. Provide the figure as a part of your submission as well. **(5 Points)**

Solution:

Code is provided in https://github.com/rahulbhadani/CPE381_FA26/blob/main/Code/HW02_ERCOT_Cubic_Spline.m

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